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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023
END SEMESTER EXAMINATION - ODD 2023
IT304 - DATA STRUCTURE AND ALGORITHM

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. The postfix form of $A*B+C/D$ is (1) CO3 BL3
 - a) $*AB/CD+$
 - b) $AB*CD/+$
 - c) $A*BC+/D$
 - d) $ABCD+/*$
2. A linear collection of data elements where the linear node is given by (1) CO3 BL2
means of pointer is called
 - a) linked list
 - b) node list
 - c) primitive list
 - d) None of these
3. Which data structure is needed to convert infix notation to postfix (1) CO3 BL2
notation?
 - a) Branch
 - b) Queue
 - c) Tree
 - d) Stack
4. How many nodes in a tree have no ancestors? (1) CO1 BL2
 - a) 0
 - b) 1
 - c) 2
 - d) n
5. Consider that n elements are to be sorted. What is the worst-case (1) CO3 BL4
time complexity of bubble sort?
 - a) $O(1)$
 - b) $O(\log_2 n)$
 - c) $O(n)$

d) $O(n^2)$

6. A full binary tree with n leaves contains

(1) CO1 BL3

- a) n nodes.
- b) $\log_2 n$ nodes.
- c) $2n - 1$ nodes.
- d) 2^n nodes.

7. The quick sort algorithm exploit _____ design technique

(1) CO3 BL2

- a) Greedy
- b) Dynamic programming
- c) Divide and Conquer
- d) Backtracking

8. The process of accessing data stored in a serial access memory is similar to manipulating data on a

(1) CO3 BL2

- a) heap
- b) queue
- c) stack
- d) binary tree

9. The smallest element of an array's index is called its

(1) CO1 BL2

- a) lower bound.
- b) upper bound.
- c) range.
- d) extraction.

10. The Postorder and Inorder traversal of a tree are as follows:

(1) CO3 BL3

Postorder: C D B E A

Inorder: C B D A E

What will be the Preorder tree?

- a) A B C D E
- b) B A D C E
- c) A B D C E
- d) A B D E C

11. Which data structure is used for implementing recursion?

(1) CO1 BL3

- a) Queue
- b) Stack
- c) Array
- d) Linked List

12. Using division method, in a given hash table of size 15, the key of value 169 be placed at position _____

(1) CO4 BL4

- a) 12
- b) 2
- c) 15
- d) 4

(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

- 13a. What is a self-referential structure? (2) CO3 BL2
- 13b. What is the difference between Union & Structure? (3) CO1 BL2
- 14a. What is ADT? (2) CO1 BL1
- 14b. Draw the expression tree: $(P-Q) / R * S + T / U$ (3) CO3 BL3
15. Convert the following Infix expression to equivalent Postfix expression: (5) CO4 BL4
 $A + (B * C - (D / E \uparrow F) * G) * H$
16. Draw the Multiway Threaded Binary tree whose sequential representation is given by: (5) CO3 BL4

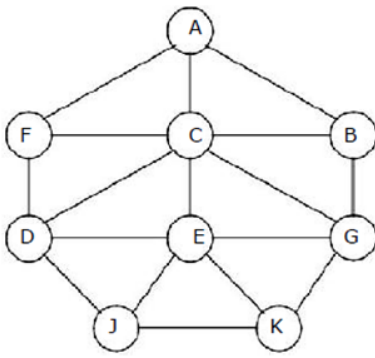
A	B	C	NULL	D	G	H	NULL	NULL	E	F	NULL	I	J	K
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17. Construct an AVL tree using the following list of integers: 12, 11, 13, 10, 09, 15, 14, 18, 7, 6, 5, 4. Show all the steps. (5) CO2 BL3

Group C

(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

- 18a. What is a binary tree? (2) CO1 BL1
- 18b. Sort the numbers using bubble sort algorithm : 12, 34, 11, 19, 5, 13 (3) CO4 BL2
- 18c. Analyze the Time Complexity and Big O notation. (10) CO1 BL2
- 19a. Construct a AVL tree from the following: (7) CO3 BL3
 21, 18, 19, 22, 28, 24, 30, 32, 27, 11 then delete 11 and 19
 Show all the steps.
- 19b. Convert the following infix expression into equivalent postfix expression using stack. (8) CO3 BL4
 $(P-Q) * R + S - T / U * V - W + X / Y$
- 20a. If N_0 be the total no of leaf nodes and N_2 be the no of nodes having two children in a binary tree, then prove that $N_2 = N_0 - 1$. (3) CO3 BL2
- 20b. Show the stages in growth of 5 order B-tree when the following keys are inserted in the given order: (7) CO3 BL4
 16, 20, 22, 42, 12, 30, 32, 18, 10, 34, 36, 38, 14, 24, 28, 40, 26
- 20c. Write an algorithm to insert an element in Circular Queue. (5) CO2 BL2
21. Answer the following: (15)
- a) Define graph. Explain its sequential and linked list representation in memory. (7) CO3 BL2
- b) Find the BFS and DFS traversal of the following graph: (8) CO3 BL3



22. Write short note on any three.

(15)

a) Output Restricted De-queue

(5) CO1 BL1

b) Prim's algorithm

(5) CO2 BL3

c) Quick Sort

(5) CO1 BL3

d) Threaded Binary Tree

(5) CO2 BL2

e) Tower of Hanoi

(5) CO1 BL1